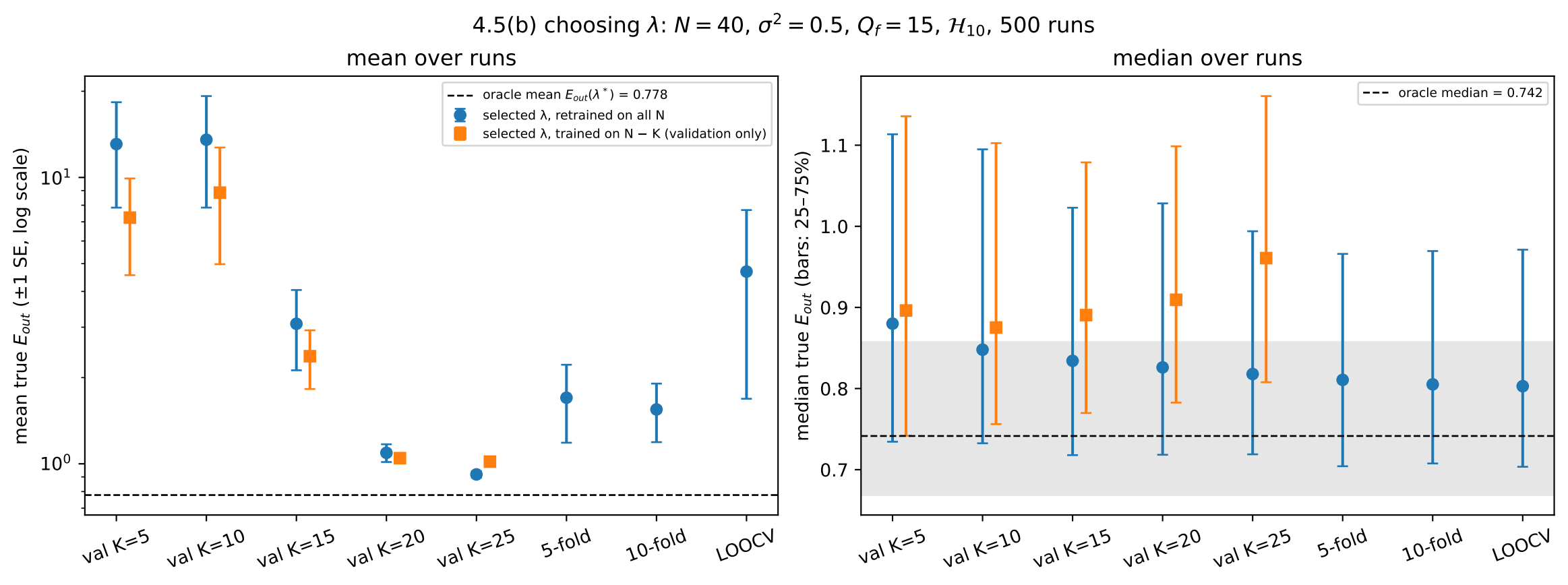
4.5(b) choosing λ: $N = 40$, $\sigma^2 = 0.5$, $Q_f = 15$, $\mathcal{H}_{10}$, 500 runs
mean over runs
median over runs
oracle mean $E_{out}(\lambda^*) = 0.778$
selected λ, retrained on all N
selected λ, trained on N − K (validation only)
oracle median = 0.742
mean true E_{out} (±1 SE, log scale)
median true E_{out} (bars: 25–75%)
10^1
10^0
1.1
1.0
0.9
0.8
0.7
val K=5
val K=10
val K=15
val K=20
val K=25
5-fold
10-fold
LOOCV